

A PRACTICAL REFERENCE

Landing Page SEO Handbook

Everything that makes a landing page rank — the technical foundation, the on-page craft, structured data, performance, and the content strategy that actually moves the needle.

Framework-agnostic. Examples use Next.js where code helps, but every principle applies to any stack.

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PART 1

How Search Actually Works

Before any tactic makes sense, you need the pipeline. Most SEO failures are not ranking problems — they are crawling or indexing problems wearing a disguise.

The four stages

Every page passes through four distinct stages. A failure at any stage means the later ones never run. Diagnosing SEO means identifying which stage you are stuck at.

1. CRAWL	A bot requests the URL and downloads raw HTML. Controlled by robots.txt, server response codes, and internal links pointing at it.
2. RENDER	The bot executes JavaScript in a headless browser to produce the final DOM. Queued separately from crawling and may lag by days.
3. INDEX	Content is parsed, deduplicated, understood, and stored. Google decides here whether the page is worth keeping at all.
4. RANK	For each query, the index is ordered by relevance, quality, authority, and user-experience signals.

Crawling is not indexing is not ranking

These three words are used interchangeably in casual conversation and they mean completely different things. Getting them straight fixes most confused SEO thinking.

STAGE	QUESTION IT ANSWERS	HOW YOU CONTROL IT
Crawling	May the bot fetch this URL?	robots.txt, server status codes, internal links
Indexing	Should this page be stored and listed?	noindex meta tag, canonical tags, content quality
Ranking	Where does it place for a query?	relevance, content depth, authority, performance

The classic mistake

Blocking a page in robots.txt to keep it out of Google. It does the opposite of what you want: the bot never fetches the page, so it never sees your noindex tag, and the URL can still be listed — as a bare link with no description — because other sites point at it. To remove a page, ALLOW crawling and serve a noindex tag.

The three pillars

PILLAR	QUESTION	NATURE OF THE WORK
Technical	Can machines access and understand it?	Mostly a one-time engineering job. Finite and completable.
On-page	Is this the best answer to the query?	Ongoing craft. Scales with how many topics you cover well.
Off-page	Does anyone else vouch for you?	Slowest. Earned over months through reputation and relationships.

Technical SEO makes you eligible to compete. It does not make you win. Teams routinely over-invest here because it is the part engineers can control, then wonder why rankings do not move. Finish it, then move on.

PART 2

Technical Foundation

Rendering strategy, crawl directives, URL hygiene, and redirects. This is the layer that decides whether your content is visible at all.

Rendering strategies

How and when your HTML gets built is the single biggest technical decision affecting SEO.

STRATEGY	HTML BUILT	CRAWLER SEES	USE FOR
CSR	In the browser	Empty shell + JS bundle	Dashboards, apps behind login
SSR	Server, every request	Complete HTML	Personalised or per-request content
SSG	Once, at build time	Complete HTML	Marketing pages, docs, blogs
ISR	Build, then on a timer	Complete HTML	CMS-backed content that changes

SSR and SSG are equivalent for SEO

A crawler cannot distinguish them. Both deliver finished HTML in the first response and there is no signal indicating which produced it. SSG simply arrives faster, which helps Core Web Vitals. For a landing page, SSG or ISR is almost always correct — SSR adds server cost and latency for no SEO gain.

The rule: if content must exist before JavaScript runs, do not use CSR. Google does render JavaScript, but rendering is queued and can lag. Every other crawler — social preview bots, AI engines, most SEO tools — reads only the raw HTML and never runs your JS at all.

robots.txt

A plain text file at the domain root. The first thing most crawlers request. It controls crawling only — never indexing.

ROBOTS.TXT

```
User-agent: *
Allow: /
Disallow: /admin/
Disallow: /*?sort=      # wildcard: any URL containing ?sort=
Disallow: /*.pdf$       # $ anchors the end of the URL

User-agent: GPTBot      # per-bot rules override the * block
Disallow: /

Sitemap: https://example.com/sitemap.xml
```

- Never disallow your JS and CSS bundles. Google needs them to render the page; blocking them makes your site look broken to the crawler.
- A missing robots.txt is fine — it means 'crawl everything'. An incorrect one is far worse than none.
- It is a public file. Do not list secret paths in it; you are advertising them to anyone who looks.
- Its main practical job on a small site is advertising your sitemap.

XML sitemaps

A machine-readable list of URLs you want indexed. It aids discovery — it does not guarantee indexing, and it does not improve rankings.

SITEMAP.XML

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url>
    <loc>https://example.com/services/mobile-apps</loc>
    <lastmod>2026-08-01</lastmod>
  </url>
</urlset>
```

- Include only canonical, indexable, 200-status URLs. Never list redirects, 404s, or noindexed pages.
- Keep lastmod honest. Faking freshness across every URL teaches crawlers to ignore the field entirely.
- Split into multiple sitemaps with an index file above 50,000 URLs or 50 MB.
- Low value on a small site. Essential once you have dozens of pages or pages that are poorly linked internally.

Canonical URLs

A canonical tag declares the single official address for a page. Without it, the same content reachable at several URLs splits its ranking signals across all of them.

IN <HEAD>

```
<link rel="canonical" href="https://example.com/pricing" />
```

The same page is commonly reachable at all of these — each is a distinct URL to a search engine:

```
https://example.com/pricing      https://example.com/pricing/
http://example.com/pricing      https://www.example.com/pricing
https://example.com/pricing?ref=x https://example.com/Pricing
```

- Always use absolute URLs in canonical tags.
- Every page should have a self-referencing canonical by default.
- A canonical is a hint, not a directive. Google may override it if your signals contradict each other.
- Pair it with 301 redirects that enforce one host and one trailing-slash convention at the server level.

Redirects and status codes

CODE	MEANING	SEO EFFECT
200	OK	Indexable. What you want.
301	Moved permanently	Passes ranking signals. Use for permanent moves.
302	Found / temporary	Signals stay with the old URL. Rarely what you want.
308	Permanent, method preserved	Like 301 but keeps POST as POST.
404	Not found	Dropped from the index over time. Correct for genuinely gone pages.
410	Gone	Like 404 but deliberate. Deindexed faster.
5xx	Server error	Serious. Repeated 5xx reduces crawl rate.

- Avoid redirect chains. Each hop costs crawl budget and dilutes signals. Redirect straight to the final destination.
- Never mass-redirect removed pages to the homepage. Google treats that as a soft 404. Redirect to the closest relevant page or serve a real 404.

URL structure

DO

/services/mobile-app-development

Short, lowercase, hyphenated, descriptive, stable, and logically nested.

AVOID

/p?id=4823&cat=7

Opaque parameters, no keywords, unstable, and often duplicated across sort orders.

- Hyphens separate words. Underscores do not.
- Lowercase only — URLs are case-sensitive and /Pricing and /pricing are two pages.
- Pick a trailing-slash convention and enforce it with redirects.
- Changing URLs costs rankings even with perfect redirects. Design the structure once, early.

HTTPS, mobile, and internationalisation

- HTTPS is a confirmed ranking signal and a prerequisite for trust. Non-negotiable.
- Google indexes the mobile version of your site. If content is hidden or missing on mobile, it is missing from the index.
- For multi-language sites, use hreflang to map equivalents. Every variant must link to every other, including itself.

```
<link rel="alternate" hreflang="en-us" href="https://ex.com/us/" />
<link rel="alternate" hreflang="en-gb" href="https://ex.com/uk/" />
<link rel="alternate" hreflang="x-default" href="https://ex.com/" />
```

PART 3

On-Page Craft

Titles, headings, copy, links, and images. The layer with the highest ratio of effort to reward, and the one most often done carelessly.

The title tag

The single highest-leverage line on any page. It is your headline in the search results and the primary driver of click-through rate.

RULE	DETAIL
Length	Roughly 50-60 characters. Google truncates by pixel width (~600px), not character count.
Order	Lead with the primary keyword. Brand name last, after a pipe or dash.
Uniqueness	Every page gets its own. Duplicate titles across pages waste them.
Honesty	It must match the page. Mismatch increases bounce, which hurts.

DO

Mobile App Development Services | Acme

Specific, front-loads the keyword, brand at the end, fits the display width.

AVOID

Home | Welcome to Acme - Best Software Company Ever Since 2011

Wasted words, brand first, truncated, no keyword intent.

Meta description

Not a ranking factor. Google has said so repeatedly and often rewrites it anyway. It matters because it is your advertising copy in the results page and directly affects click-through rate.

- Aim for 150-160 characters.
- Write it as a sales pitch with a clear benefit, not a summary.
- Include the keyword — matched terms are bolded in results, which draws the eye.

The meta keywords tag is dead

Google stopped using it in 2009 and has publicly confirmed it is ignored. It does nothing. Every other major engine ignores it too. If you see it in a template, it is a fossil.

Heading hierarchy

- Exactly one H1 per page, describing the page's single subject.
- H2 for major sections, H3 for subsections. Never skip levels for visual reasons — style with CSS instead.
- Headings are a document outline, not decoration. Screen readers and crawlers both navigate by them.
- Put real keywords in them naturally. 'Our Approach' tells a crawler nothing; 'Our Mobile App Development Process' does.

Content depth and search intent

Every query has an intent behind it. One page serves one intent well. This is the single most important structural insight in SEO.

INTENT	EXAMPLE QUERY	PAGE THAT SERVES IT
Informational	what is an ERP system	Guide, article, glossary entry
Commercial	best software agency	Comparison, case studies, service page
Transactional	hire react developers	Service page with clear CTA
Navigational	acme login	The specific page they want

Why one page cannot rank for everything

A single URL realistically ranks for one tight cluster of related queries. Targeting 'mobile apps', 'AI development', 'ERP' and 'SEO services' from one page means each dilutes the others and none ranks. Split them: one URL per intent, each with its own H1, copy, and title, all linked from a hub page.

- Depth beats repetition. Keyword stuffing stopped working over a decade ago and now reads as spam to both readers and classifiers.
- Cover the topic properly — answer the obvious follow-up questions on the same page.
- Use natural related vocabulary. Modern engines match meaning, not exact strings.

Internal linking

- Links pass authority between your own pages. A page with no internal links pointing at it is nearly invisible.
- Anchor text tells the engine what the destination is about. 'Click here' wastes the signal; 'mobile app development services' spends it well.
- Keep important pages within three clicks of the homepage.
- Link contextually from body copy, not only from nav and footer.

Image SEO

- Descriptive alt text on every meaningful image. Empty alt (alt="") for purely decorative ones so screen readers skip them.
- Descriptive filenames: mobile-app-team.webp, not IMG_4821.png.
- Serve modern formats — WebP or AVIF — and compress at source. A 3 MB hero image is a Core Web Vitals failure regardless of tooling.
- Always set explicit width and height, or reserve the space with CSS. Missing dimensions are the top cause of layout shift.
- Lazy-load below-the-fold images. Never lazy-load the hero — it is usually your LCP element and lazy-loading it directly delays LCP.

PART 4

Structured Data

How to state, in machine-readable terms, what your page and your organisation actually are — rather than hoping the crawler infers it.

What it is and why it matters

Prose says you build software. Structured data states that you are a `ProfessionalService` with a founder, a service catalogue, and a service area. The vocabulary is `schema.org`; the format is JSON-LD, a script tag in your head that users never see.

Set expectations correctly

Google is explicit that structured data is not a direct ranking boost. What it does: makes you eligible for rich results, which lift click-through; establishes you as an entity in the knowledge graph; and feeds AI answer engines, which increasingly cite structured sources.

The types that matter for landing pages

TYPE	USE IT FOR
Organization	Any company. Name, logo, contact, social profiles.
LocalBusiness	Businesses with a physical location. Requires a real address.
ProfessionalService	Agencies and consultancies. Subtype of <code>LocalBusiness</code> .
Service / OfferCatalog	The individual services you sell.
Product + Offer	Anything with a price. Enables price and availability in results.
FAQPage	Question-and-answer blocks. Can expand your listing.
BreadcrumbList	Navigation path. Replaces the raw URL in results.
Article / BlogPosting	Editorial content, with author and dates.
Person	Founders, authors. Feeds E-E-A-T.
WebSite	The site itself; enables a sitelinks search box.

Writing it: use @graph

Emit one connected graph rather than several disconnected blocks. The `@id` cross-references let a crawler resolve everything into a single entity instead of unrelated islands.

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@graph": [
    {
      "@type": ["Organization", "ProfessionalService"],
      "@id": "https://ex.com/#org",
      "name": "Acme", "url": "https://ex.com",
      "logo": { "@id": "https://ex.com/#logo" },
      "founder": { "@id": "https://ex.com/#founder" },
      "areaServed": "Worldwide",
      "sameAs": ["https://linkedin.com/company/acme"]
    },
    { "@type": "Person", "@id": "https://ex.com/#founder",
      "name": "Jane Doe", "worksFor": { "@id": "https://ex.com/#org" } }
  ]
}
</script>
```

Rules that keep you out of trouble

- Only describe content that is actually visible on the page. Marking up services or reviews that appear nowhere is a guidelines violation.
- Never fabricate. Invented addresses, fake review counts, and made-up ratings risk a manual penalty and are indefensible if audited.
- sameAs is how engines confirm you are one entity across the web. Point it at real, active profiles.
- Keep name, address, and phone identical everywhere — site, Google Business Profile, directories. Inconsistency actively suppresses local rankings.
- Validate after every change with the Rich Results Test and validator.schema.org. They catch different classes of error.

PART 5

Performance and Core Web Vitals

Three measured signals, taken from real visitors rather than a lab. A real if modest ranking factor — and a large conversion factor.

The three metrics

METRIC	MEASURES	GOOD	USUAL CAUSE WHEN BAD
LCP	Time until the largest element is visible	< 2.5s	Heavy images, slow server, render-blocking resources, JS-gated content
INP	Responsiveness to taps and clicks	< 200ms	Long JavaScript tasks blocking the main thread
CLS	How much layout jumps while loading	< 0.1	Images without dimensions, injected banners, late web fonts

INP replaced First Input Delay in March 2024. If a guide still talks about FID, it is out of date.

The animation trap

The most common self-inflicted LCP wound

Scroll-reveal libraries start elements at opacity:0. When the page is server-rendered, that invisible state is baked into the delivered HTML. The text is present and crawlable, but nothing is visible until the JS bundle downloads, hydrates, and flips it — so your LCP is delayed by the entire cost of hydration, and the page is blank if JS fails.

The fix is to animate with CSS so the first paint is already meaningful, and to respect reduced-motion preferences:

CSS-DRIVEN REVEAL — WORKS WITHOUT JS

```
@media (prefers-reduced-motion: no-preference) {
  .reveal { animation: fade-up .6s ease-out both; }
}
@keyframes fade-up {
  from { opacity: 0; transform: translateY(20px); }
  to   { opacity: 1; transform: none; }
}
```

Never apply reveal animations to above-the-fold content. The hero should paint immediately; it is almost always the LCP element.

Practical performance rules

- Compress images at source. No framework optimiser makes a 3 MB PNG acceptable as a source asset.
- Set fetchpriority="high" on the hero image and preload it. Let everything else lazy-load.
- Use font-display: swap and preload your primary font. Late fonts cause both layout shift and invisible text.
- Subset fonts to the characters you use. Full multi-language font files are frequently larger than all your images combined.
- Ship less JavaScript. Every kilobyte is parse time on the main thread, which is what INP measures.
- Self-host fonts and critical assets. Third-party origins add DNS, TLS, and unpredictable latency.

Lab data versus field data

Lighthouse gives you lab data — one simulated load on one machine. Google ranks on field data: the Chrome User Experience Report, gathered from real visitors over 28 days. A perfect Lighthouse score with poor field data means your real users are having a worse time than your test machine. Always read the field section first.

PART 6

Social Sharing and Metadata

How your link looks when a human pastes it into Slack, LinkedIn, or WhatsApp. Not a ranking factor — a click-through factor.

Open Graph and Twitter Cards

```
<meta property="og:type"          content="website" />
<meta property="og:url"          content="https://ex.com/" />
<meta property="og:title"        content="Page title" />
<meta property="og:description"  content="One-line pitch." />
<meta property="og:image"        content="https://ex.com/og.png" />
<meta property="og:image:width"  content="1200" />
<meta property="og:image:height" content="630" />
<meta property="og:site_name"    content="Acme" />

<meta name="twitter:card"  content="summary_large_image" />
<meta name="twitter:title" content="Page title" />
```

- The image must be 1200x630 pixels for the large-card format. Wrong dimensions get cropped badly or rejected.
- Use an absolute URL for og:image. Relative paths silently fail on most platforms.
- Declaring summary_large_image without providing an image gives you a bare, unattractive card — worse than declaring nothing.
- Most scrapers do not run JavaScript. These tags must be in the server-rendered HTML.
- Platforms cache aggressively. Use their debuggers to force a refresh after changes.

PART 7

Content and Site Architecture

Where the real, durable traffic comes from — and the part no amount of technical polish can substitute for.

One intent, one URL

This is the structural decision that determines your ceiling. A single-page site can rank for one thing. A well-structured site can rank for hundreds.

A STRUCTURE THAT CAN RANK

/	-> brand + primary positioning
/services	-> hub linking to every service
/services/mobile-apps	-> one intent, own H1 and title
/services/ai-development	-> one intent, own H1 and title
/case-studies/acme-erp	-> proof, with real numbers
/blog/what-is-an-erp	-> informational, captures early intent

Topic clusters

Group related content around a comprehensive pillar page. The pillar targets the broad competitive term; each cluster page targets a specific long-tail query. They link to each other, concentrating authority on the pillar.

- Long-tail queries are less competitive and convert better. Ten pages ranking for specific queries beat one page ranking for nothing.
- Every cluster page links up to the pillar; the pillar links down to all of them.

Keyword research, briefly

- Start from what customers actually say, not what you call it internally. Sales calls and support tickets are the best source.
- Check real search volume and difficulty before committing. Free options: Google Search Console for what you already surface for, Keyword Planner, and autocomplete.
- Read the current top ten results. They tell you what Google believes the intent is — match that format or lose.
- Target realistic difficulty early. Ranking eleventh for a huge term earns nothing; ranking first for a specific one earns customers.

E-E-A-T

Experience, Expertise, Authoritativeness, Trustworthiness. Google's quality framework. Not a single score, but a set of signals human raters and algorithms both look for.

- Named authors with real, verifiable credentials — not 'Admin'.
- Genuine case studies with specific numbers and named clients.
- Visible contact details, a real address where applicable, and clear ownership information.
- Citations to primary sources; original data or research where you have it.

On fabricated testimonials

Invented reviews are a legal exposure in many jurisdictions, a manual-penalty risk if marked up as structured data, and trivially detectable by prospects who look for the companies named. Placeholder testimonials must be replaced with real ones before a site takes traffic.

Off-page: links and reputation

- Backlinks remain the strongest off-page signal. Quality vastly outweighs quantity — one link from a respected industry publication beats a hundred directory listings.
- Earn them: original research, genuinely useful free tools, open-source work, conference talks, guest articles, and being quoted as a source.
- Never buy links. It violates Google's guidelines and the penalties are severe and slow to lift.
- For local businesses: a claimed Google Business Profile with consistent name, address, and phone across every directory usually outperforms any on-site change.

PART 8

Measurement and Tooling

You cannot optimise what you do not measure. These are the tools that tell you the truth, and what each is actually for.

The essential set

TOOL	WHAT IT ANSWERS
Google Search Console	What queries you surface for, your position, and every indexing error. Free and non-optional.
Rich Results Test	Does my structured data parse, and what rich results do I qualify for?
validator.schema.org	Is my schema vocabulary correct? Stricter than Google's tool.
PageSpeed Insights	Core Web Vitals, both lab and real-user field data.
Bing Webmaster Tools	Same as Search Console for Bing — which also feeds some AI engines.
curl	What does the raw HTML actually contain before any JavaScript?

The single most useful check

```
curl -s https://example.com | grep -o "Your headline text"
```

If that prints, your content is in the first response and no crawler needs to run JavaScript to see it. If it does not print, nothing else in this handbook will save you.

What to actually track

- Impressions before clicks. Impressions tell you whether you are visible at all; clicks tell you whether your title is working.
- Average position per query, not sitewide averages, which are meaningless.
- Indexed page count versus submitted. A large gap means Google is rejecting pages, and Search Console will say why.
- Core Web Vitals field data, not lab scores.

Be patient and change one thing at a time

SEO changes take weeks to show effect. Recrawling, reindexing, and re-ranking all take time. Changing ten things at once means you learn nothing about which one worked. Make a change, note the date, wait, then read the data.

PART 9

Checklists

Print these. Run the launch checklist before every deploy and the audit quarterly.

Pre-launch: technical

- ☐ Content visible in raw HTML (verify with curl, not the browser inspector).
- ☐ Built and tested against a production build — never the dev server.
- ☐ Unique title tag under 60 characters on every page.
- ☐ Unique meta description of 150-160 characters on every page.
- ☐ Exactly one H1 per page; heading levels never skipped.
- ☐ Self-referencing canonical tag on every page.
- ☐ robots.txt resolves, allows crawling, and does not block JS or CSS.
- ☐ sitemap.xml resolves, lists only canonical 200-status URLs.
- ☐ HTTPS enforced; one host and one trailing-slash convention via 301s.
- ☐ Custom 404 page that returns an actual 404 status code.

Pre-launch: content and metadata

- ☐ Every meaningful image has descriptive alt text; decorative ones have empty alt.
- ☐ All images compressed and served as WebP or AVIF, with width and height set.
- ☐ Open Graph tags complete, including a real 1200x630 image at an absolute URL.
- ☐ JSON-LD present, valid, and describing only visible content.
- ☐ Internal links use descriptive anchor text.
- ☐ No placeholder or fabricated content anywhere — testimonials especially.
- ☐ No dead anchors or links pointing at unbuilt sections.
- ☐ Mobile layout verified on a real device, not just a resized window.

Post-launch: within the first week

- ☐ Domain verified in Google Search Console and Bing Webmaster Tools.
- ☐ Sitemap submitted; URL Inspection run on the homepage.
- ☐ Rich Results Test passed on every page type.
- ☐ PageSpeed Insights run; field data reviewed once it accumulates.
- ☐ Social preview checked by pasting the link into Slack and LinkedIn.
- ☐ Analytics installed and recording.

Common mistakes to check for

MISTAKE	WHY IT HURTS
Blocking JS/CSS in robots.txt	Google cannot render the page and sees it as broken.
Using robots.txt to deindex	The bot never sees your noindex tag; the URL can still be listed.
opacity:0 on above-the-fold content	Delays LCP by the full hydration cost; blank page if JS fails.
One page targeting many keywords	Each dilutes the others; none ranks.

Lazy-loading the hero image	Directly delays the LCP element.
Duplicate titles across pages	Wastes the highest-leverage line you have.
Redirect chains	Wastes crawl budget and dilutes signals at each hop.
Redirecting dead pages to home	Treated as a soft 404 and ignored.
Missing image dimensions	The leading cause of layout shift.
Keyword stuffing	Ineffective for over a decade; reads as spam.

The order of operations

If you remember nothing else, remember the sequence. Technical work is finite and makes you eligible to compete. Content and links are unbounded and are what actually win.

PHASE	WORK	TIMESCALE
1. Foundation	Rendering, robots, sitemap, canonicals, metadata, schema	Days
2. Performance	Core Web Vitals, images, fonts, JavaScript weight	Days to weeks
3. Structure	One intent per URL, hub pages, internal linking	Weeks
4. Content	Cluster pages, case studies, genuinely useful articles	Months, ongoing
5. Authority	Backlinks, reputation, citations, profiles	Months, ongoing

The honest summary

Phases 1 and 2 are perhaps a week of focused engineering, and they are the part most teams over-invest in because it is the part engineers control. They close every technical gap and then stop paying dividends. Phases 3 to 5 have no end and are where essentially all durable organic traffic comes from. Finish the first two quickly, then spend your attention on the rest.